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Experiencing Music

The concept of music is frequently introduced as "organized sound." Encyclopedia Britannica supports this idea, explaining that while any sound could potentially be musical, each culture establishes boundaries for what its musicians consider acceptable. Michael Linton builds upon this, proposing that music is "the organization of sound and silence into forms that carry culturally derived meanings, cultivated for aesthetic or utilitarian purposes." This act of human organization appears central to our understanding of music, setting it apart from naturally occurring "musical" phenomena like the sounds of waterfalls or birds, which we typically don't classify as "music." From an aesthetic standpoint, music is seen as a skillful or pleasing arrangement of sound and silence. Nattiez (1990) offers a relativistic view, stating that music is whatever a society deems it to be, while noise is what is considered disruptive or unpleasant.

The late 1800s marked the beginning of scientific inquiry into the relationship between sound and how we perceive it. During this time, several definitions framed music as a form of communication, capable of transmitting moods, emotions, thoughts, impressions, or even philosophical, sexual, or political ideas. This communicative capacity has led to the description of music as a "universal language." In contrast, Levi R Bryant suggests that musical experience functions not as a language but as a system of signs used for problem-solving, similar to mathematics. Nevertheless, the significant complexity found within music's structural elements lends credence to the idea of music as a language. For example, musical genres are distinguished by their unique methods of articulating, organizing, and disseminating sound and silence, creating a system with a level of intricacy comparable to that of 'language'.

Music as a subjective experience

The notion that music ought to be "pleasant" (judged esthetically) or "melodic" (assessed neutrally and/or esthetically) is frequently used to contend that mere sound organization isn't sufficient for something to be music. Because the boundaries of what constitutes music are fluid across cultures and time, more comprehensive definitions incorporate the cultural and social evolution of music. These definitions accept that while specific interpretations may vary, certain universal aspects persist. The evolving perception of dissonance, once considered noise, illustrates this shifting divide. Luciano Berio highlighted how the Tristan chord, now a recognized harmonic structure, was once deemed noise in the 19th century due to its deviation from established harmonic conventions.

However, a subjective understanding of music extends beyond traditional ideas of pleasantness or melody. Luciano Berio proposed that music is "everything one listens to with the intention of listening to music." This definition centers on the listener's experience rather than the inherent structure of the sound. Consequently, "found" sound structures—whether originating from natural phenomena or algorithmic processes—can be classified as music if they are engaged with through the aesthetic cognitive processes associated with music appreciation. This perspective allows the distinction between music and noise to be dynamic, evolving with cultural interpretations over time, differing between

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contemporary cultures, and varying among individuals based on their unique experiences and inclinations.

Cultural context plays a significant role in determining what is perceived as music versus noise or unpleasant sounds. What is appreciated as music in Bali, for example, might be dismissed as simply "a din" by a British listener. Similarly, the historical era shapes musical perception. What is considered acceptable music in the West today would likely have been ridiculed in the 17th century.

Music Cognition

The cognitive definition offers a perspective where music transcends the physical sound and its immediate perception. It proposes that music functions as a crucial mechanism through which our brains organize perception, guide action, and structure memory.¹ This definition holds significant sway in cognitive science, where researchers seek to map the neural correlates of musical experience, specifically identifying brain regions involved in parsing and recalling its various components.

Clifton's phenomenological definition

In his 1983 work, Thomas Clifton defines music as "an ordered arrangement of sounds and silences whose meaning is presentative rather than denotative," underscoring its intrinsic value. He expands on this by saying that "music is the actualization of the possibility of any sound whatever to present to some human being a meaning which he experiences with his body—that is to say, with his mind, his feelings, his senses, his will, and his metabolism" (Clifton 1983). Therefore, music, according to Clifton, is "a certain reciprocal relation established between a person, his behavior, and a sounding object" (Clifton 1983, 10).

Clifton's distinction between music and non-music hinges on human behavior rather than compositional methods or the physical characteristics of sound. He defines "musical behavior" as a state where "one whose very being is absorbed in the significance of the sounds being experienced." This goes beyond simply listening; it encompasses perceiving, interpreting, judging, and feeling. The musically behaving person experiences musical significance *through* the sounds, not just by hearing them.

Clifton identifies two primary distinctions between music and non-music: (1) the presentative nature of musical meaning and (2) the level of personal involvement. Significantly, his definition deliberately avoids aesthetic judgments. He argues that music isn't a fixed entity but a meaning constructed by individuals. To understand this experience meaningfully, Clifton suggests we must be open to the composition's own order and significance, question our assumptions about musical materials, and recognize the value in articulating this meaningful experience (Clifton 1983).

Xenakis's definition

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Composer Iannis Xenakis, in his work "Towards a Metamusic," offers a multifaceted definition of music, outlining seven key aspects:

First, he states that music is a fundamental *comportment* or way of being for anyone who thinks and creates it. Second, it represents an *individual realization*, a personal manifestation. Third, Xenakis sees music as the *fixing in sound of imagined virtualities*, encompassing cosmological, philosophical, and other intellectual arguments. Fourth, he posits that music is *normative*, acting unconsciously as a model for how to be or how to act through a form of sympathetic influence. Fifth, music is *catalytic*, its very presence allowing for internal psychological or mental shifts, much like the effect of a hypnotist's crystal ball. Sixth, he describes it as the *gratuitous play of a child*, suggesting an element of freedom and inherent joy. Finally, Xenakis characterizes music as a *mystical (but atheistic) asceticism*, implying a form of discipline and transcendence that goes beyond typical religious connotations. Consequently, expressions of common emotions like sadness, joy, or love, and dramatic scenarios, are considered only very limited and specific instances within this broader definition.

Musical Analysis

The thoughtful integration of music into daily life not only enriches our experiences but also offers a holistic pathway to overall well-being. Consequently, the ability to analyze music is a fundamental skill for all music therapists, as crucial as understanding a client's musical preferences – their likes and dislikes for specific melodies, songs, or musical systems.

While personal musical preferences often lack explicit reasoning, they are frequently rooted in mental associations with particular genres or melodies, leading to liking, disliking, or neutrality. Despite the seemingly irrational basis of these preferences, music therapists must meticulously analyze the genres favored by their clients. Understanding the client's current emotional state is equally vital. If the therapeutic goal involves mood alteration, the music therapist must carefully select genres with the potential to facilitate the desired emotional shift. This is where musical analysis becomes an indispensable skill. The effective application of music therapy is a journey of continuous experimentation with music and clients, requiring long-term dedication. This makes music therapy a demanding yet rewarding field for aspiring practitioners, necessitating significant effort and commitment to achieve expertise.

For music therapists, a comprehensive understanding of musical forms and their impact on individuals is paramount. This demands a deep knowledge of diverse musical genres and systems, coupled with rigorous research and analysis to identify the most appropriate music for each client's unique needs. Given music's demonstrated benefits for many, an in-depth understanding of its effects is essential.

Emerging evidence suggests that music can be a valuable tool in overcoming various physical, mental, and spiritual challenges encountered throughout the human lifespan – from birth to growth and death. Music can serve as a supportive companion through life's inevitable struggles. Recent research further indicates that the conscious application of music can positively influence one's attitude and approach to life, facilitating smoother

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resolution of conflicts and promoting inner balance, mirroring the inherent equilibrium found within music itself.

Music itself encompasses a spectrum of emotions and experiences – highs and lows, sadness and joy, helplessness and empowerment, conflict and peace – which alternate and expand, ultimately resolving in a balanced and often pleasant manner. This inherent balancing act within music forms the philosophical basis for its adoption as a constant companion in life. All musical forms possess the capacity to harmonize conflicts and to do so in a way that is often pleasant. By engaging with music, the mind can develop a remarkable resilience to face and cope with life's adversities.

Therefore, music can be effectively employed in addressing the needs of individuals across the lifespan and in various challenging situations – premature infants, children, adolescents, adults, the middle-aged, the elderly, the sick, and those facing helplessness or unbearable circumstances. Fundamentally, music aids in coping with difficulties, fostering courage and fortitude in the face of adversity, and instilling new hope. It has the power to transform hopelessness into hope, depression into confidence, and tension into light-heartedness – provided we are open to listening and embracing change.

Music is rapidly emerging as a promising "future medicine" with potential curative applications in numerous fields, including geriatrics, child development, stress management, pain management, mental and community health, elder care, attitudinal development, and palliative care. The potential applications are vast.

However, for those unfamiliar with it, music can seem like an incomprehensible jumble. Despite its profound impact, the precise mechanisms by which music addresses human challenges and facilitates positive behavioral changes remain largely unclear.

How does such a transformation take place?

The question of why we have specific musical tastes and responses is currently challenging to answer, as it is intricately linked to the human brain, an organ that remains as enigmatic to scientists as music itself. The neurological basis for individual preferences and varying reactions to music is yet to be fully understood. The age-old question of differing tastes remains unanswered in this domain. However, this lack of a clear explanation does not diminish the undeniable impact of music.

Our present understanding allows us to appreciate the extensive and profound avenues for expression inherent in music, along with the significant creativity it entails. While sound in itself requires no justification, the human mind is constantly seeking meaning and purpose, often constructing it even in its absence. Subjective responses can be contextualized by studying the reactions of homogeneous groups to particular musical genres.

A musical composition often reflects a deeply personal expression from the composer, effectively conveying their resolutions without needing logical arguments. These expressions can range from the accessible nature of folk music to the complex structures of classical traditions. The key is the listener's engagement and interest in the composer's artistic

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communication. Ultimately, the value lies in the listener's subjective experience of the music, surpassing any rational critical analysis.

The process by which music affects us, though seemingly straightforward, is highly complex, involving our brain's intricate responses. Despite modern scientific advancements, our understanding of the brain remains limited. Consequently, a complete understanding of music's interaction with the brain is currently unattainable.

However, the very mystery that envelops music adds to its allure. Roger Scruton wisely noted that intellectual analysis alone cannot capture the essence of musical experience: "An expert musicologist may not understand what he hears, despite his skilled descriptions. The decisive fact is the experience itself." Engaging directly with music is essential to explore its potential and to pave the way for new applications that benefit humanity.

Pulsating Music

Since the earliest times, humanity has recognized the connection between consciousness and *nada* (intonation) for health and well-being, even though the precise mechanisms by which sound influences consciousness remain a mystery to this day. The alignment of sound with consciousness has been celebrated throughout history through communal singing, dancing, drumming, and other practices. It has long been understood that the quality of sound and rhythm can affect one's state of consciousness.

A fundamental healing principle of sound and music lies in the power of sound vibration. Throughout ancient history, sound vibrations have been utilized as therapeutic agents. Even the most primitive societies demonstrated an awareness of a specific mental state – a healing consciousness – in which individuals recovered from their ailments. These communities employed instruments like drums and cymbals, as well as vocalizations such as whistles, murmurs, and shouts, to induce a trance-like state in the patient. There has been a persistent belief that the human body is a living entity composed of vibrations and wavelengths. It is the vibration within music that facilitates connection through its pulsations, initiating a similar vibration in another body and triggering a response of a similar frequency.

Involvement of Mind in Music

Our entire musical experience inherently involves our mind. The mind shapes music, and in turn, is shaped by it. Because music and the mind share similar structures or patterns, music can be seen as analogous to the human psyche. It serves as a deeply felt reflection of our innermost being, where life's accumulated disappointments, regrets, and traumas become layered over time. When music engages with these deeply held experiences, it performs a kind of alchemy: pain transforms into pleasure, agony into ecstasy, and melancholy into a sweet tenderness. This process can foster a newfound confidence to navigate life's challenges with music as a supportive companion.

Music, Mind and Meaning

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While sound itself is self-contained, the human mind is inherently drawn to understanding its sense or meaning. Because a musical composition typically reflects a composer's internalized theme, it is the resolutions they communicate – the artistic outcomes, rather than logical justifications – that capture the listener's imagination. These resolutions can be expressed directly (as in folk music) or with great complexity (as in orchestral music), but the ultimate goal is to instill a sense of motivation within the listener.

Analytical and Emotional Mind Reflected in Music

Rhythmic time structures exist in both music and the human mind. Furthermore, the analytical and emotional aspects of the brain (left and right hemispheres) are reflected in music. Achieving a balance between analysis and emotion (intuition) is crucial for daily life, as an excess of either can be detrimental. Ancient music practices aimed at cultivating this balance. Thus, integrating music into daily life can promote a healthy equilibrium between analytical and emotional inputs, potentially serving as a form of meditative practice.

The classical music systems of the West and India illustrate this balancing act beautifully. Indian *ragas*, while deeply emotional, are always performed with a keen awareness of subtle rhythms (*laya*). When practicing compositions like *varnams* or *kritis*, the emotional expression of the *raga* is interwoven with precise adherence to the *tala* (beat). This simultaneous engagement of the right brain with emotion and the left brain with rhythmic calculation is a remarkable "acrobatic" feat in music.

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Emotion-rich Beats, Rhythms and Tones

It is a fundamental aspect of composition that composers manipulate rhythmic and melodic structures to evoke intended emotions in their audience. A compelling illustration is Pyotr Ilyich Tchaikovsky's iconic Op. 31 Slavonic March. According to the composer, its debut in Moscow in 1876 unleashed a "whole storm of patriotic enthusiasm." In this work, Tchaikovsky strategically employed authentic Serbian folk songs to imbue a local character and heighten its emotional content.

A closer examination reveals how the tempo and pacing contribute to a feeling of hardship. The seemingly unsteady rhythms, which undeniably amplify the emotion, are intricately combined with a clear, calculated melody built from precise tones, adding to the overall emotional effect. The entire piece is constructed around an elaborate system of "analytical" melodies in its polyphonic and contrapuntal textures, juxtaposed with "intuitive" or "emotional" beats characterized by their seemingly wavering tempo. This masterful combination allows listeners to vividly imagine a wounded yet resilient soldier attempting to rise from a war-ravaged landscape and slowly make their way towards home, their unsteady progress reflected in the music's gait.

Music and intuition

We have seen how music mirrors both the scope and the profundity of our mental faculties. Intuition, that inner flash of understanding or insight, lies outside the domain of analysis, surpassing all argument and discussion. It is fundamentally rooted in one's own subjective experience.

Indian philosophy regards knowledge derived solely from sensory perception as "false knowledge." Conversely, intuition occupies the highest position in Indian thought, considered "right knowledge" or the "supreme knowledge," often described as the eternal, boundless knowledge of truth. Ancient wisdom held that intellectual development devoid of intuition could lead to societal harm due to its inherent incompleteness. It was also believed that music offered a path to transcend the limitations of purely intellectual growth, as genuine cultural progress was attributed to the cultivation of intuition, not just analytical intelligence.

Music that Triggers the Psychic Process

Since ancient times, music and rhythms have been deliberately used in rites and rituals to elevate individuals' consciousness to higher planes of existence. Echoing this, Lehtonen (1986) noted that musical experience is a powerful activator of psychic processes. Ancient systems like shamanism, religious chanting, African drumming rituals, and Nada Yoga have all celebrated music within their distinctive practices, employing its tones, lilts, beats, and tunes specifically to transport consciousness.

Music and Time Experience

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Plato defined time as "the moving image of eternity," while Kant viewed time and space as fundamental "forms of intuition" that structure our experience, placing every object within a temporal sequence. Consequently, our perception of sensations and objects is inherently linked to a chronological order.

The proper understanding of a musical piece hinges on our capacity to engage with its temporal dimensions: past, present, and future. As Susanne Langer eloquently put it, "music is time made audible." This temporal sense in music arises from tonal shifts (in frequency or micro-intervals) and the creation and release of rhythmic tension.

Even without musical training, listeners perceive music as a fluid continuum connecting past, present, and future, revealing a point where the "timeless" intersects with the temporal. Music, therefore, exists within time and allows us to value the present moment, often overshadowed by mental activity. The "living-in-now" experience often induced by music can temporarily quiet the analytical mind, potentially explaining its use as a pain reliever.

Neurological studies have shown that musical timing can alter brainwave patterns. The tempo and beat frequency of music can induce either relaxation or stimulation. Low-pitched, slow music tends to relax listeners, whereas high-pitched, fast music is known to be activating, possibly due to increased beta wave activity in the brain.

The Mental Impact of Pitch

The effect of pitch on human consciousness is an ancient understanding, predating even spoken language in musical cultures. It has long been observed that low pitches resonate with deeper, heavier emotions, while high pitches are linked to lighter feelings and a sense of detachment. The binding quality of low pitch is often seen as intuitive and spiritual, explaining its widespread use in chants and mantras across religions and by spiritual guides. These low sounds can provide serenity amidst worldly stress and potentially lead a dedicated listener to profound inner depths, where ego dissolves into universal love and compassion.

In contrast, high pitch tends to elicit lighter, "feel good" emotions like joy, fostering confidence and assurance in facing the world. It can also sharpen the mind, promoting alertness and a detached objectivity that may cultivate independence. However, high pitch is also associated with intense emotions such as anger and frustration, suggesting its potential for anger management and calming agitation.

Notably, high-pitched notes are consistently found in music intended to energize or stimulate listeners, often in conjunction with fast tempos and intricate rhythms.

The Healing Tonic

The Himalayan singing bowls and Tibetan trumpets generate tones abundant in harmonics and semi-tones, promising a deeply resonant auditory experience. Concentrating on these sounds is believed to alter one's state of consciousness. The expanding harmonics from a singing bowl can be beautifully imagined as a lotus flower unfurling, representing the hope

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and optimism of embracing another bright day. Many who have experienced these demonstrations have confirmed the wave-like emanation of harmonics when the singing bowl is activated.

Ups and Downs: The Ladders and Snakes in Music

The ups and downs of life find their sonic parallel in ascending and descending musical scales. Similarly, the vertical structures of Western music, like chords and counterpoint, can create a feeling of the communal support that accompanies us even in life's loneliest passages.

Music also incorporates unexpected twists, turns, surprises, and novelties, mirroring the shocks, agonies, ecstasies, thrills, and traumas we inevitably face in our life journey.

The regular coming and going of notes or melodies can be seen as an auditory reflection of the precious people who join our lives and eventually depart, leaving behind cherished memories.

However, the underlying continuity of notes and phrases, despite all these variations, reminds us to maintain optimism and to approach life with confidence and courage.

The built-in Tensions and their Resolutions in Music

Igor Stravinsky's (1973) description of music as a series of tensions seeking relaxation resonates with all forms of musical expression. A fundamental characteristic of music is its underlying pattern of continuously emanating tension and resolution.

The opening line of Mahatma Gandhi's bhajan, 'Hare Rama, Hare Rama, Rama Rama, Hare Hare', illustrates this point. The initial repetition of 'Hare Rama' can be heard as a tension that resolves in the subsequent 'Rama Rama Hare Hare', creating a musical question and answer. This inherent dialogue is present in all music, progressing until a harmonious conclusion is reached.

Whether folk or classical, ethnic or sophisticated, all musical forms exhibit this tension-resolution pattern, readily apparent to the attentive listener. This "dialogue" can even become a vibrant "quarrel" between instrumental voices, as seen in Hindustani *jugalbandhis*, which ultimately resolve in a joyful manner.

This fundamental question-answer or tension-resolution structure is universal across global musical traditions. Interestingly, our own breathing cycle mirrors this pattern: inhalation creates bodily tension, while exhalation brings release. Recent studies indicate that the musical build-up and release of tension are encoded as electrical pulses in the brain.

Sigmund Freud noted our innate capacity to derive intense pleasure from contrast, a quality abundantly provided by music through its interplay of sound and silence, notes and pauses, and various rhythmic and tonal variations.

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Intriguingly, the rhythms of nature also display similar patterns in timing (*tala*), intensity, synchronicity, and frequency contrasts (*swaras*).

Neurological research by Dr. Gordon Shaw and his associates has revealed that specific brain cell firing patterns resemble the structural organization of Baroque, New Age, and Eastern music.

Analogous to how muscles gain flexibility through repeated tension and relaxation in exercise, the mind benefits from the recurring tension and resolution within musical pieces, ultimately enhancing its ability to cope with environmental stresses and traumas.

Musical Process: Striving towards Balance and Harmony

Music, in its essence, seeks equilibrium amidst its inherent contrasts, be they melodic contours, dynamic shifts, or even musical arguments. The *alap* in Indian music exemplifies this, where musicians improvise to navigate the unexplored paths of *ragas*.

A key aspect of Indian music is the strict adherence to *sruti*, a fundamental discipline. Simultaneously, it offers immense freedom for creative musicians to discover new melodic heights and depths, varied tempos and dynamics, and diverse ornamentation styles. The inherent tonal flexibility of *ragas* provides an expansive canvas for continuous innovation. The analytical exploration that initiates a *raga alap* often blossoms into intuitive understanding.

Listening attentively to a Carnatic *raga* like Dharmavati, we can discern profound emotional undertones, even as the *raga* meanders through upward and downward phrases anchored by the Sa-Pa-Sa drone of the *tanpura*. The unwavering purity of the *sruti* provided by the drone constantly reminds both listener and performer to remain centered and balanced, despite the melodic ascents and descents.

Consequently, music achieves a balance between emotion and intellect, ensuring neither is diminished. This balance fosters a state of equanimity or mental equilibrium – a highly regarded mental state in Indian spiritual traditions.

Musical Peace

The ultimate aim of all religious traditions is *Shanti*, or peace. We turn to music seeking inner peace and well-being. Saint Thyagaraja, a prolific composer who enriched *raga* experience, believed peace was essential for well-being. Personal experience in workshops has shown the profound peace and harmony evoked by slow *raga* renditions and even Western classical pieces like the Blue Danube.

The increasing significance of music as a non-invasive, holistic medicine highlights its potential as a dependable complementary system for alleviating physical and psychological issues. Modern societal trends – migration, urbanization, globalization, competition, disconnected brain activities, and accelerated lifestyles – have created a pace of life that

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outstrips our natural biological rhythms. This has led to social alienation, fractured families, rising frustration and depression, isolation, loss of identity, loneliness, and trauma, leaving individuals feeling marginalized and robotic in a mechanized world. The weakening of emotional bonds has increased susceptibility to stress.

Music offers a vital refuge, serving as a non-invasive means to address stress, depression, anger, and social disharmony. Scientists and philosophers now acknowledge the historical contribution of diverse musical systems in maintaining ethical behavior and social norms across all societies.

However, increased population displacement and migration are significantly altering traditional regional musical expressions, leading to new genres that diverge from past harmonies. In this changing landscape, a scientific understanding of music's often-hidden nuances is increasingly important for its evolution to meet individual and social needs. Music can awaken dormant aspects of our being. Let us trust in the power of music.